Prediction based on sufficient dimension reduction.

23.01.2018

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Joint work with

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Outline

1. Introduction
2. Sufficient Dimension Reduction (SDR)
3. The model
4. Results
5. Example

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- Objective in Statistics: Given a sample $(X_1, Y_1), \ldots, (X_n, Y_n)$ we want to estimate $r(x)$.
  - Consistency and Asymptotic Distribution.
  - Confidence Intervals for $r(x)$.
  - Prediction.

Australian Athletes example

A sample of $n = 202$ individuals training at the Australian Institute of Sport.

Objective

Explain and Predict the (LBM) Lean Body Mass (Y) as a function of the 6 predictive variables $X = (W, Z)$

W indicator for gender ($W = 0$ for males and $W = 1$ for females)

Z logarithms of

- height
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Problem

The principal components reduction does not take into account the response of interest.
Important information on the relation between Y and X may be lost, affecting future predictions.

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No information about the regression is lost in reducing the dimension of X if the reduction is sufficient.

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The distribution of $Y|X$ is the same as the distribution of $Y|\beta^T X$.

The span of β is a dimension reduction subspace.

β is 1-dimensional, so the structural dimension of this regression is $d = 1$.

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- Inverse Reduction based on models: it is based on specify a parametric model for the inverse regression of X on Y. Under this approach the SDR, $R(X)$ is not necesary linear on X. (Cook (2007), Cook and Forzani (2008))

Model for the inverse regression

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One of the most remarkable characteristics of this approach is that given a model for $X|Y$ can itself be inverted to provide a method for estimating the forward mean function $E(Y|X)$ without specifying a model for the full joint distribution of (X, Y), (Adragni and Cook (2009)).

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This fact suggests an estimation of the regression $r(x)$ as

$$\widehat{r}_n(x) = \sum_{i=1}^{n} w_i(x) Y_i \quad \text{where} \quad w_i(x) = \frac{\hat{f}(x|Y_i)}{\sum_{i=1}^{n} \hat{f}(x|Y_i)}.$$

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where $f(R(x)|Y)$ denotes now the density of $R(X)|Y$, evaluated at $R(x)$.

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Then we can consider an estimator of the regression $r(x)$ as

$$\widehat{r}_n^{\text{res}}(x) = \sum_{i=1}^{n} w_i^{\text{res}}(x) Y_i \quad \text{with} \quad w_i^{\text{res}}(x) = \frac{\hat{f}(\hat{R}(x)|Y_i)}{\sum_{i=1}^{n} \hat{f}(\hat{R}(x)|Y_i)}, \tag{2}$$

where $\hat{f}(\cdot|y)$ denotes an estimator of the conditional density of $R(X)|Y = y$ and $\hat{R}(x)$ is the estimated reduction.

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- the mean $\eta_y = \mu + \beta\nu_y$ depends linearly on ν_y,
- the distribution of $X|Y = y$ is a particular case of exponential family f_{η_y}.

Generalized (inverse) linear model for (X, Y)

Australian Athletes example

$Y =$ Lean Body Mass (LBM)

$X = (W, Z)$

W indicator $W = 0$ (males) and $W = 1$ (females)
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We can not assume that $X|Y$ is normal!!

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where

- $T(x) = (t_1(x), \ldots, t_k(x))^T$ is a vector of known real-valued functions,
- $\eta \in \mathbb{R}^k$ is the vector of natural parameters.
- $h(x) \geq 0$ is a non-negative known function,

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Reduced Rank:
$\mathrm{rank}(\beta) = d$, $\beta = AB$ with $A \in \mathbb{R}^{k_1 \times d}$ semi-orthognal and $B \in \mathbb{R}^{d \times r}$.

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Reduced Rank Generalized Linear Model (RRGLM)

SDR under RRGLM

Under RRGLM

- $X|(Y=y) \sim f_{\eta_y}(\cdot)$

$$f_{\eta_y}(x) \quad = \quad \exp\{\eta_{y1}^T T_1(x) + \eta_{y2}^T T_2(x) - \psi(\eta_y)\} h(x)$$

- *Linear*:

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Theorem: (Bura et al. 2016) $A^T T_1(X)$ is a Sufficient Dimension Reduction.

Regression Estimation under inverse models

If we assume that $X|Y = y \sim f(\cdot|y, \xi_0)$ the representation of the regression function indicates that

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This expression of the regression function suggests that it can be estimated through a plug-in procedure, replacing the true parameter ξ_0 with any consistent estimator $\widehat{\xi}_n$

$$\widehat{r}_n(x) = \frac{\sum_{i=1}^n Y_i f(x|Y_i, \widehat{\xi}_n)}{\sum_{i=1}^n f(x|Y_i, \widehat{\xi}_n)} = \sum_{i=1}^n w_i(x) Y_i,$$

with $w_i(x) = \frac{f(x|Y_i, \widehat{\xi}_n)}{\sum_{i=1}^n f(x|Y_i, \widehat{\xi}_n)}$.

Regression Estimation under inverse models based on SDR

If we assume that $X|Y = y \sim f(\cdot|y, \xi_0)$ under a reduce rank generalized linear model (RRGLM), then $R(X) = A^T T_1(X)$ is a sufficient dimension reduction (Bura et al. (2016)).

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Then, the representation of the regression function indicates that

$$r(x) = \frac{E\{Yf(A^T T_1(x)|Y, \xi_0)\}}{E\{f(A^T T_1(x)|Y, \xi_0)\}}$$

Regression Estimation under inverse models based on SDR

The regression function under the RRGLM can be estimated with

$$\widehat{r}_n^{\text{res}}(x) = \sum_{i=1}^{n} w_i^{\text{res}}(x) Y_i,$$

where

$$w_i^{\text{res}}(x) = \frac{f(\hat{A}^T T_1(x) | Y_i, \widehat{\xi}_n^{\text{res}})}{\sum_{i=1}^{n} f(\hat{A}^T T_1(x) | Y_i, \widehat{\xi}_n^{\text{res}})},$$

where $\widehat{\xi}_n^{\text{res}}$ denote an estimator of ξ_0 under the rank restriction model.

Regression Estimation under inverse models

- Under GLM: the true parameter ξ_0 with any consistent estimator $\widehat{\xi}_n$

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$$\sqrt{n}\{\widehat{r}_n(x) - r(x)\}$$

is asymptotically normal with mean zero and asymptotic variance

$$avar(\sqrt{n}\{\widehat{r}_n(x) - r(x)\}) = C^{-2} E([\{Y - r(x)\} g_{0,x,\xi_0}(Y) + DL(X, Y)]^2).$$

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The asymptotic results for $\widehat{r}_n(x)$ and $\widehat{r}_n^{\text{res}}(x)$ assume that

- ξ_0 is consistently estimated by $\widehat{\xi}_n$, and with $\widehat{\xi}_n^{\text{res}}$ in the restricted rank model.
- the inputted estimators of ξ_0 satisfies a linear expansion.

In the case, when the MLE these assumptions are guaranteed (Bura et al. (2016)).

Asymtotic Results

Under RRGLM:

If $R(X)$ is a sufficient dimension reduction for the regression of Y on X. Then, for any y_1 and y_2, we have that

$$\frac{f(x|y_1)}{f(x|y_2)} = \frac{f(R(x)|y_1)}{f(R(x)|y_2)},$$

therefore

$$\widehat{r}_n^{\text{res}}(x) = \frac{\sum_{i=1}^n Y_i f(x|Y_i, \widehat{\xi}_n^{\text{res}})}{\sum_{i=1}^n f(x|Y_i, \widehat{\xi}_n^{\text{res}})}.$$

In practice, to use the estimator $\widehat{r}_n^{\text{res}}(x)$ we do not need to choose $\hat{R}(X)$.

Example

Australian Athletes example: A sample of $n = 202$ individuals training at the Australian Institute of Sport.

Objective

Explain and Predict the (LBM) Lean Body Mass (Y) as a function of the 6 predictive variables $X = (W, Z)$

W indicator for gender ($W = 0$ for males and $W = 1$ for females)

Z logarithms of

- height
- weight
- red cell count
- white cell count
- hemoglobin

Example: Prediction and Confidence intervals

The distribution of $X \mid Y$ follows an exponential family. Bura et al. (2016) obtained a sufficient reduction for the regression of $Y|X$ is $R(x) = \beta^T(w, z^T)^T$.

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A plug-in procedure can be used to estimate the asymptotic variance

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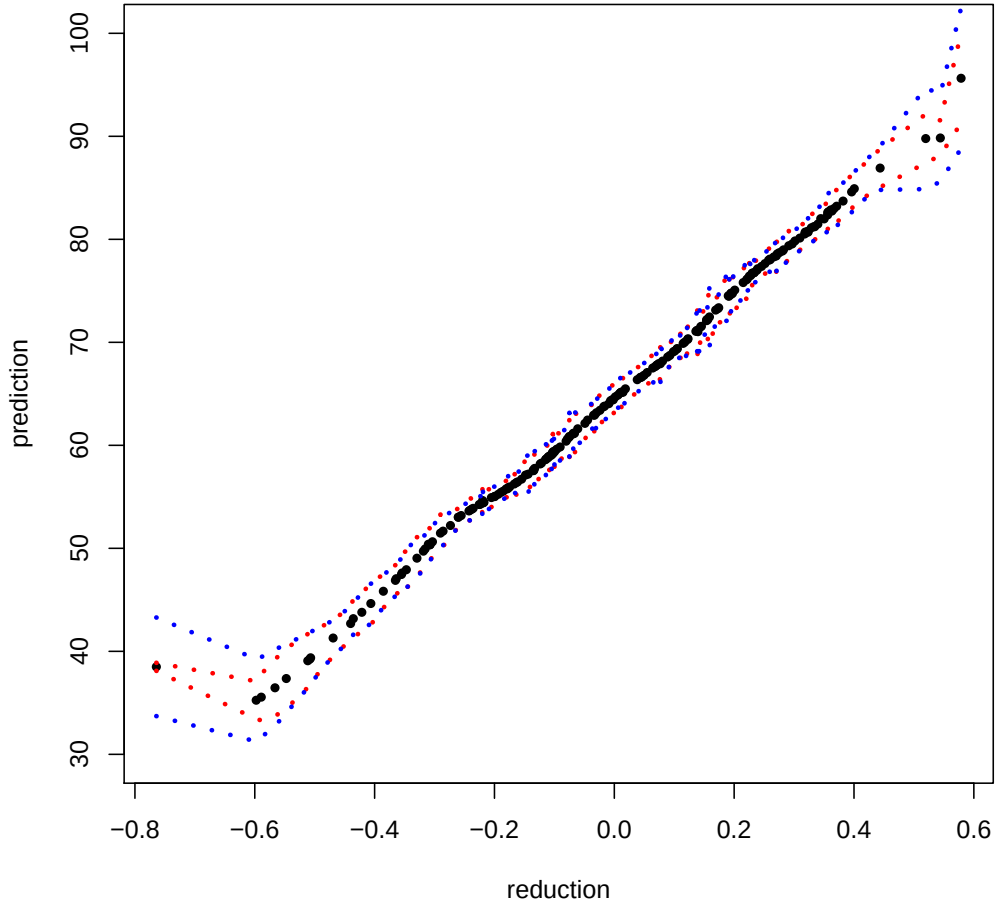
A bootstrap variance estimation can be computed

$$\mathrm{IC}_{Boot}(r(x)) = \widehat{r}_n(x) \pm z_{\alpha/2}\sqrt{\mathrm{var}_{Boot}\{\widehat{r}_n(x)\}}.$$

Example: Confidence intervals

LBM prediction versus estimated sufficient reduction. Confidence intervals for $r(x)$: asymptotic in red and bootstrap in blue.



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It may also be desirable to construct prediction intervals for the response at a fixed value x of the predictors.

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$$F_{Y|X=x}(y) = E\{I_{(Y \leq y)} \mid X = x\} = \frac{E\{I_{(Y \leq y)} f(x|Y)\}}{E\{f(x|Y)\}}.$$

Example: : Prediction intervals

Mimicking the construction of the estimators of $r(x)$, we arrive at the following estimators of $F_{Y|X=x}(y)$

$$\begin{aligned}
\widehat{F}_{Y|X=x}(y) &= \frac{\sum_{i=1}^{n} I_{(Y_i \leq y)} f(x|Y_i, \widehat{\xi}_n)}{\sum_{i=1}^{n} f(x|Y_i, \widehat{\xi}_n)} \\
\widehat{F}^{\text{res}}_{Y|X=x}(y) &= \frac{\sum_{i=1}^{n} I_{(Y_i \leq y)} f(x|Y_i, \widehat{\xi}^{\text{res}}_n)}{\sum_{i=1}^{n} f(x|Y_i, \widehat{\xi}^{\text{res}}_n)}.
\end{aligned}$$

and we derive the their asymptotic properties.

Example: : Prediction intervals

For the Australian Athletes example and for each observation i, $1 \leq i \leq 202$, we computed a 95% prediction interval considering $X_{new} = X_i$ and using the 201 observations (X_j, Y_j) with $j \neq i$.

Example: : Prediction intervals

For the Australian Athletes example and for each observation i, $1 \leq i \leq 202$, we computed a 95% prediction interval considering $X_{new} = X_i$ and using the 201 observations (X_j, Y_j) with $j \neq i$.

In 95.05% of the cases, the constructed intervals contain the observed response Y_i.

Conclusion and Further work

- We have proved the consistency and asymptotic normality of the prediction based on a sufficient dimension reduction when a generalized linear model is postulated for the inverse regression.

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- We illustrated the results computing confidence intervals for the regression function.
- We also presented prediction intervals, estimating $F_{Y|X=x}$.

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We are thinking in some extensions of these results.

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We are thinking in some extensions of these results.

- We are developing the prediction procedure without specified a parametric model

- We are studing how to explote the estimation of $F_{Y|X=x}$ in order to estimate any parameter associated with the conditional distribution defined through a continuous functional.

Thank you for your time and attention!

Daniela Rodriguez
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